Bell-State Preparation on a Two-Legged Ladder QPU

Dynamic swap, and unitary GHZ-uncompute

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The Challenge

Goal. Prepare a Bell state

$$\left|\Phi^+\right\rangle_{e_0 e_1} = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$$

between the two *far-end* qubits of a two-legged ladder QPU, for any ladder length L.

Constraints.

- Only nearest-neighbour gates (ladder edges).
- Top leg, bottom leg, rungs all allowed.
- Must scale to arbitrary L.

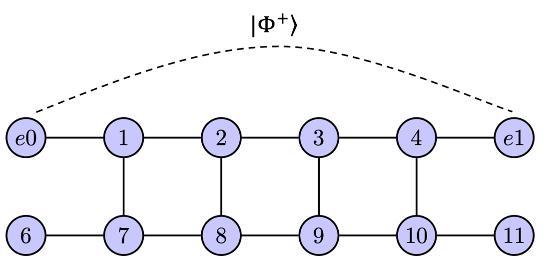

Figure 1: The challenge example: $L = 5$ (12 qubits).

Our Protocols

1. Entangement Swapping with Feed-Forward

Uses alternating Bell pairs, Bell measurements on intermediate qubits, and a final classically controlled Pauli correction to deterministically entangle the endpoints.

Benefit: Constant quantum depth independent of ladder length.

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2. Middle-Out GHZ-Uncompute (MOGU)

Builds a GHZ state from the middle of the top leg outward, then uncomputes the intermediate qubits to leave only the endpoints in a Bell state.

Benefit: Measurement-free operation with optimal linear depth.

Motivation 1: Entanglement swapping

Start with two Bell pairs $(0,1)$, $(2,3)$ — Bell-measure the middle pair $(1,3)$ — Pauli-correct 0 or 3 based on the outcome — $0,3$ end up in $|\Phi^+\rangle$ though they never interacted.

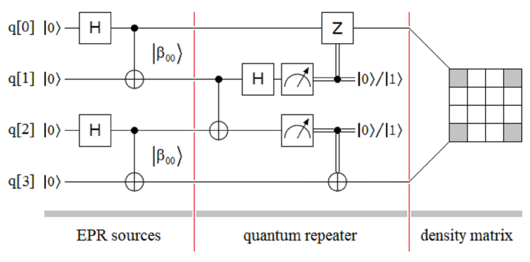
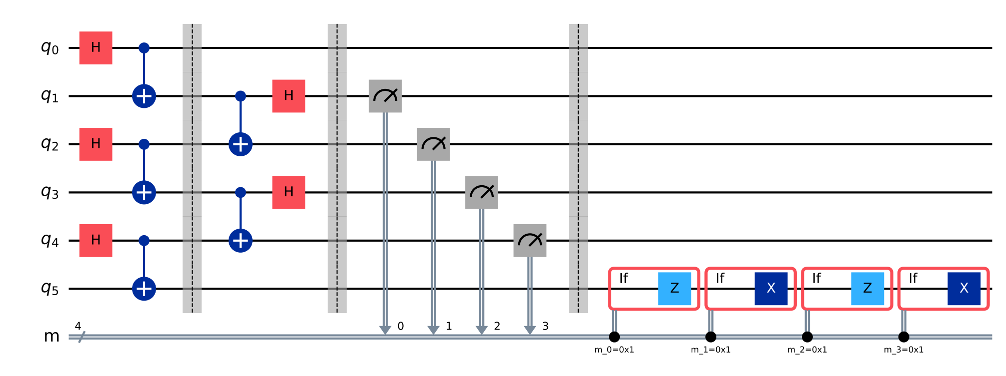


Figure 2: Quantum circuit for entanglement swapping

Mastriani, *Sci. Rep.* 13, 21998 (2023)

Protocol I — Entanglement-swap with feed-forward

Three layers, $O(1)$ depth: (1) three Bell pairs on alternating links; (2) two parallel Bell-measurements on the inner pairs; (3) classically-controlled Pauli correction on e_1.



Protocol I — The Core Math: Stabilizer

We represent quantum state with stabilizer.

The quantum state is the eigenstates of all stabilizer with $+1$ as eigenvalue.

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Example 1: $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ can be represented by two stabilizer:

$$\boxed{X_1X_2,\ Z_1Z_2}$$

Example 2: $|000000\rangle$ can be represented by six stabilizer:

$$\boxed{Z_1,\ Z_2,\ Z_3,\ Z_4,\ Z_5,\ Z_6}$$

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$$H_i X_i H_i = Z_i, \; H_i Z_i H_i = X_i \tag{1}$$

$$\Rightarrow Z_i \longrightarrow X_i, \quad X_i \longrightarrow Z_i \tag{2}$$

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$$X_c \longrightarrow X_c X_t, \quad X_t \longrightarrow X_t \tag{3}$$
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We used these rules to derive the Stabilizer for the final state.

Protocol I — The Core Math: Results

After the Bell-measurement layer, the residual stabilisers on (e_0, e_1) reduce to

$$\tilde{P}_{XX} = (-1)^{m_1+m_3}\, X_{e_0} X_{e_1}, \qquad \tilde{P}_{ZZ} = (-1)^{m_2+m_4}\, Z_{e_0} Z_{e_1}.$$

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Feed-forward Pauli correction on e_1:

$$\boxed{X^{m_2 \oplus m_4}\ Z^{m_1 \oplus m_3}}$$

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Generalisation

- Even N: induction on Bell-pair count $\Rightarrow$ depth ≤ 7 for all L.
- Odd N: a single GHZ-3 link at the tail handles the parity obstruction.

Resources. $N{+}1$ or $N{+}2$ two-qubit gates, N measurements, constant depth.

Motivation 2: GHZ-uncompute

Idea. If you cannot measure, build the entanglement everywhere and then carefully *uncompute* what you don't want.

1. Create GHZ state $\frac{1}{\sqrt{2}}(|0\cdots0\rangle + |1\cdots1\rangle)$ across the top leg.
2. Reverse the fan-out from the centre outward $\Rightarrow$ interior qubits return to $|0\rangle$; only the endpoints stay entangled.

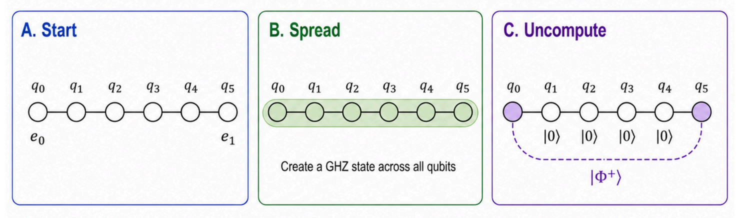
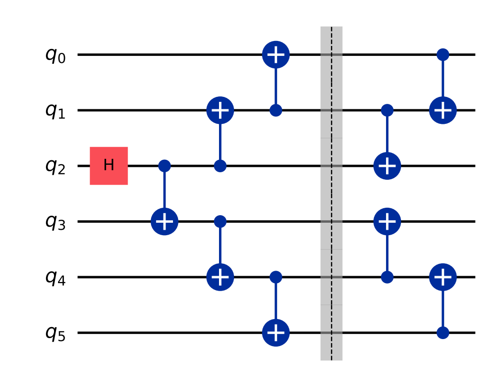


Figure 3: Create GHZ state across all qubits and then disentangle the middle qubits.

In this way, there is zero measurements. We call this **MOGU** = *Middle-Out GHZ-Uncompute*.

Protocol II — MOGU (measurement-free)

Three phases: (1) H on middle qubit $q_{\lfloor L/2 \rfloor}$; (2) spread CNOTs fan out from middle to ends; (3) shrink reverse-fan-out disentangles every interior qubit, centre outward.



Protocol II — Core formulas

Sequential schedule.

$$\text{depth} = L + 2, \quad \text{2Q gates} = 2L - 1, \quad \text{measurements} = 0.$$

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Stabiliser sketch. After Phase B the state is the standard $(L{+}1)$-qubit GHZ. Phase C peels off each interior Z_i generator, leaving

$$\langle X_{e_0} X_{e_1},\ Z_{e_0} Z_{e_1},\ Z_1, \dots, Z_{L-1} \rangle \ = \ \mathrm{stab}\big(\big|\Phi^+\big\rangle_{e_0 e_1} \otimes |0\rangle^{\otimes L-1} \big).$$

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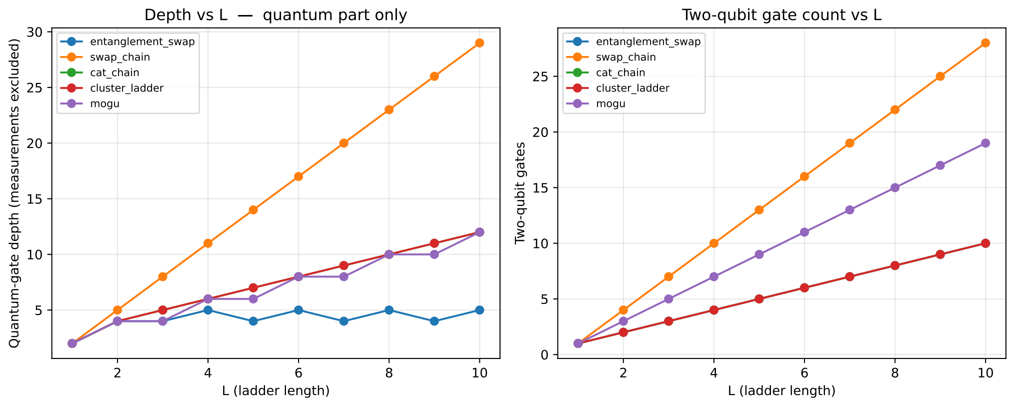
Parallel schedule (documentation-only, not in the verified code): if we interleave Phase C with the still-in-flight Phase B, the depth drops to

$$d_{\|}(L) \ \approx \ \begin{cases} \min(L+2,\ \lceil L/2 \rceil + 4) & L \text{ even}, \\ \min(L+1,\ \lceil L/2 \rceil + 3) & L \text{ odd}. \end{cases}$$

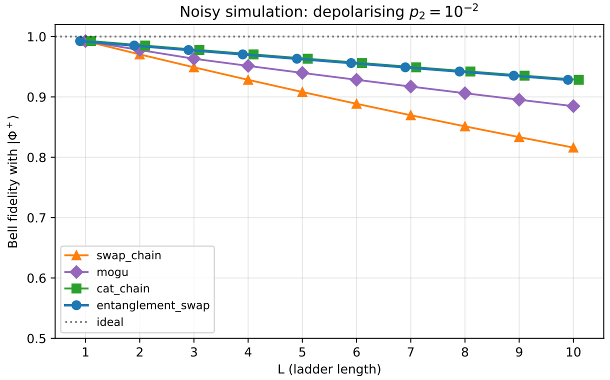
$\approx 2\times$ *depth reduction at large L — free constant-factor win.*

Benchmark: Resources vs L



What you see. Protocol I (blue) is the only depth-$O(1)$ curve — its quantum part stays at ≤ 5 layers for every L. Measurements and feed-forward Paulis are excluded from this count (they execute in classical real time). MOGU is linear with the smallest slope among the unitary protocols; SWAP-chain dominates both axes.

Benchmark: Bell fidelity under depolarising noise



Markers for entanglement_swap and cat_chain are horizontally offset by ±0.1 for visibility; fidelities coincide at every integer L.

Symmetric two-qubit depolarising error $p_2 = 10^{-2}$ on every CNOT.

At $L = 10$**:**

- Protocol I: 0.928
- cat_chain: 0.928 (same 2Q-gate count)
- MOGU: 0.885
- swap_chain: 0.816

Protocol I (•) and cat_chain (■) give *identical* fidelities at every L — both incur $\approx N$ noisy CNOTs touching (e_0, e_1). The markers are jittered by ± 0.1 in L purely for visibility.

References & Summary i

Take-home

- We proposed two protocols.
- Protocol I: $O(1)$ depth, classical feed-forward.
- Protocol II (MOGU): $L+2$ depth, measurement-free, depth-optimal for unitary 1D.

Key references

- Żukowski, Zeilinger, Horne, Ekert, *Phys. Rev. Lett.* **71**, 4287 (1993) — entanglement swapping primitive.
- Briegel, Dür, Cirac, Zoller, *Phys. Rev. Lett.* **81**, 5932 (1998) — quantum repeater.
- Bäumer *et al.*, *PRX Quantum* **5**, 030339 (2024), arXiv:2308.13065 — long-range CNOT teleportation on 101 IBM qubits.
- Song *et al.*, *Phys. Rev. Applied* **24**, 024068 (2025), arXiv:2409.06989 — constant-depth fan-out with feedforward.

Thank you.

Questions?